CV-041

Senior AI/ML Engineer

Genpact · United States · Data & AI

Skills

Python, SQL, JavaScript, TypeScript, FastAPI, REST APIs, boto3, Microservices, Event-driven architecture, PyTorch, TensorFlow, Scikit-Learn, Xgboost, Hugging Face Transformers, PEFT/LoRA, Model Distillation, CNN, Transformer Models, Computer Vision, NLP, Feature Engineering, Data Modeling, Model Evaluation, LangChain, LlamaIndex, Semantic Kernel, OpenAI API, Amazon Bedrock, Gemini, GPT-4

Work experience

Senior AI/ML Engineer · GENPACT

2023-03 – 2026-09

Architected a multi-tenant Agentic AI platform for enterprise Source-to-Pay that used hybrid retrieval over SQL and pgvector with Amazon Bedrock embeddings, enabling RAG-grounded agents to summarize intake requests and cut manual triage time by 45% across 12 client workspaces.
Implemented the hybrid retrieval layer that combined structured SQL filters with pgvector semantic search, tuning thresholds for supplier discovery while documenting precision and recall tradeoffs for retrieval quality reviews.
Designed a two-tier supplier master data architecture with global reference schemas and…

Senior AI/ML Engineer · Lyra Health

2020-09 – 2023-02

Led productionization of AI and ML solutions for oncology and telehealth on Azure, deploying CNN and Transformer models with PyTorch that reduced diagnostic turnaround by 35% and improved early detection rates across clinical workflows.
Developed multi-cancer screening models with multimodal inputs spanning imaging, labs, and pathology using PyTorch and Hugging Face Transformers, expanding product coverage and improving diagnostic accuracy for 4 clinical programs.
Fine-tuned LLMs with PEFT and LoRA and deployed asynchronous inference through Ray Serve, optimizing model performance via Ray…

Software Engineer / ML Engineer · Lyra Health

2018-10 – 2020-07

Built CNN-based computer vision models for MRI and CT scans using Faster R-CNN and Mask R-CNN, achieving a 12% improvement in tumor detection sensitivity that supported radiologist workflow automation.
Developed readmission risk prediction models on structured EHR data with scikit-learn and XGBoost, identifying high-risk patients and reducing readmissions by 18% across 2 hospital systems.
Applied NLP models with BioBERT and LSTM attention mechanisms to extract diagnoses and comorbidities from unstructured clinical notes, accelerating physician review workflows by 30%.
Built and orchestrated…

Research Engineer · Empower Health

2017-05 – 2018-08

Forecasted marketing activity performance using deep learning models for 132 companies with 90% accuracy.
Translated forecast outputs into budget allocation guidance that improved campaign ROI by 15% for enterprise healthcare clients.
Applied natural language processing methods to classify user-generated content into relevant topics, improving content categorization accuracy by 22% for marketing research workflows.
Designed automated ETL workflows for aggregating marketing and behavioral data from multiple healthcare platforms and third-party APIs.
Connected healthcare platform and…

Education

- Bachelor's Degree — Florida International University (2014–2018)

Latest: Computer Science — Florida International University